

Spirit . Intellect . Purpose



**University of
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Computer System

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Computer system



A **computer system** consists of a set of hardware and software which processes data in a meaningful way.

Computer

is a programmable machine that receives input, stores and manipulates data, and provides output in a useful format.

is a device that accepts information (in the form of digitalized data) and manipulates it for some result based on a program or sequence of instructions on how the data is to be processed

Data and Information



Computers process data into information.

Data is a collection of unprocessed items, which can include text, numbers, images, audio, and video.

Information conveys meaning and is useful to people.

Computer system



The two principal characteristics of a computer are:

- It responds to a specific set of instructions in a well defined manner.
- It can execute a prerecorded list of instructions (a program).

Modern computers are electronic and digital .

The actual machinery wires, transistors, and circuits is called **hardware**. the instructions and data are called **software**.



Hardware

are the physical components of the computer system which can be seen, felt and touched, while the software are the non-physical components.

- a. Input devices
- b. Output devices
- c. Storage devices
- d. Processing devices

Hardware



All general-purpose computers require the following hardware components:

- **Memory:** Enables a computer to store, at least temporarily, data and programs.
- **Mass storage device :**Allows a computer to permanently retain large amounts of data. Common mass storage devices include disk drives and tape drives.
- **Input device:** Usually a keyboard and mouse are the input device through which data and instructions enter a computer.

Hardware



- **Output device:** A display screen, printer, or other device that lets you see what the computer has accomplished.
- **Central processing unit (CPU):** The heart of the computer, this is the component that actually executes instructions.

In addition to these components, many others make it possible for the basic components to work together efficiently.

- For example, every computer requires a bus that transmits data from one part of the computer to another.

Software



refers to the programs, which are required to operate the computer.

a. Application software

is computer software designed to help the user to perform singular or multiple related specific tasks

Examples:

Enterprise software, Accounting software, Office suites, Graphics software and media players

b. System software

computer software designed to operate the computer hardware and to provide and maintain a platform for running application software.



Peopleware

computer operator / user

Procedure

a set of coded instructions that tell a computer how to run a program or calculation.

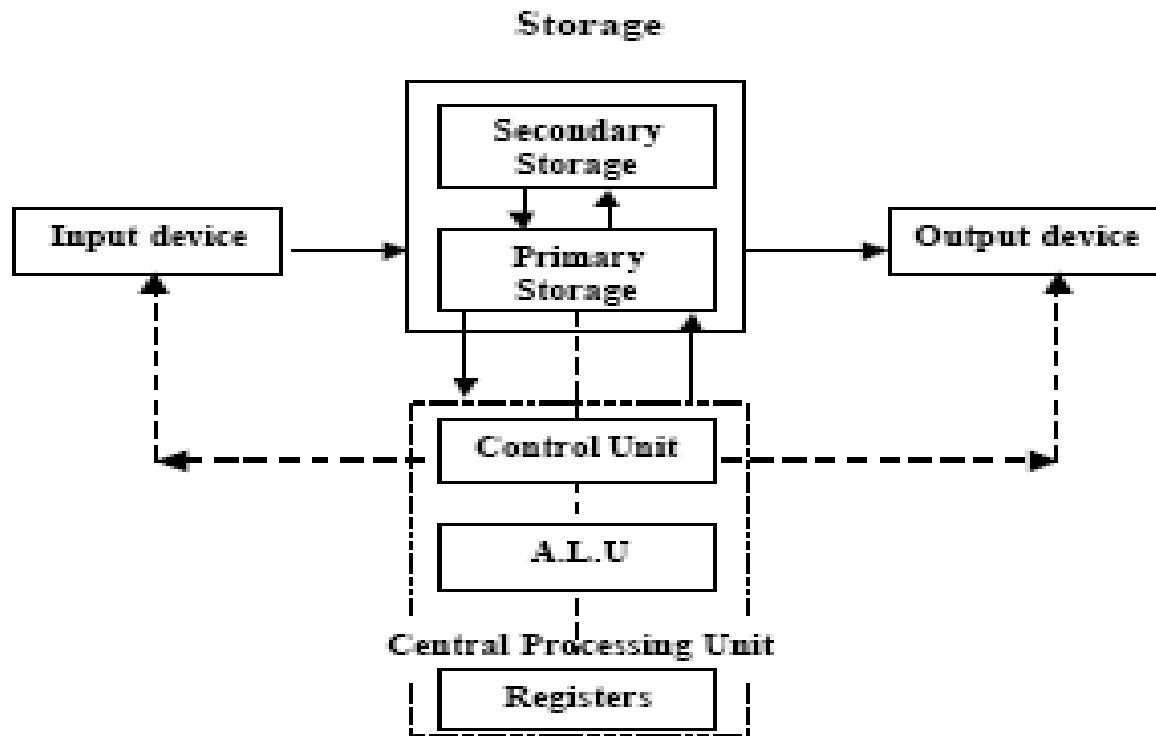
Connectivity

how hardware or software devices can communicate with other devices

COMPONENTS OF COMPUTER SYSTEMc



- Input Unit/Devices
- Output Unit/Devices
- Central Processing Unit
- Memory Unit/Storage Devices



COMPONENTS OF COMPUTER SYSTEM



The Central Processing Unit is made up of certain functional elements including:

- 1. Arithmetic Logic Unit (ALU)**
- 2. Control Unit (CU)**
- 3. Registers**
- 4. Buses**
- 5. Clock**

Arithmetic Logic Unit

The arithmetic and logic unit of CPU is responsible for all arithmetic operations like addition, subtraction, multiplication and division as well as logical operations such as less than, equal to and greater than. Actually, all calculations and comparisons are performed in the arithmetic logic unit.

COMPONENTS OF COMPUTER SYSTEM



Control Unit

The control unit is responsible for controlling the transfer of data and instructions among other units of a Computer. It is considered as the “**Central Nervous System**” of computer, as it manages and coordinates all the units of the computer. It obtains the instructions from the memory, interprets them and directs the operation of the computer. It also performs the physical data transfer between memory and the peripheral device.

COMPONENTS OF COMPUTER SYSTEM



Registers

Registers are small high-speed circuits (memory locations), which are used to store data, instructions and memory addresses (memory location numbers).

Buses

A group of wires on the main circuit board of the computer. It is a pathway for data flowing between components.

Clock

Clock is another important component of CPU, which measures and allocates a fixed time slot for processing each and every micro-operation (smallest functional operation).

Types of Bus



1. Internal bus

Also known as system bus made up of 4 distinct parts.

a. Power bus

Wires that provide electrical power to every part of the motherboard

Types of Bus



b. Control bus

A **control bus** is a part of a computer's bus system that is used to manage and coordinate the activities of the computer's various components. It carries control signals from the CPU to other parts of the computer, such as memory and input/output devices, to ensure that data is transferred correctly and operations are synchronized.

The control bus typically includes signals for:

- **Read/Write operations:** Indicating whether data is being read from or written to memory.
- **Interrupts:** Allowing devices to signal the CPU that they need attention.
- **Clock signals:** Synchronizing the timing of operations across different components.

Types of Bus



c. Address bus

An **address bus** is a collection of wires or pathways used in a computer system to transfer the address of a memory location or a device. It is a crucial component in the architecture of a computer, enabling the CPU to communicate with memory and other peripherals by specifying the location where data needs to be read from or written to.

Types of Bus



Key Points about Address Bus:

- **Function:** Carries the address information from the CPU to memory or I/O devices.
- **Unidirectional:** Typically, the address bus is unidirectional, meaning it only carries information in one direction—from the CPU to the memory or I/O devices.
- **Width:** The width of the address bus (number of lines) determines the maximum addressing capacity of the system. For example, a 32-bit address bus can address (2^{32}) (4,294,967,296) unique memory locations.

Types of Bus



Example:

Consider a computer with a 16-bit address bus. This means the address bus can carry 16 bits of information, allowing the CPU to address up to (2^{16}) (65,536) unique memory locations. If the CPU wants to read data from memory location 0x1A3F, it will place the address 0x1A3F on the address bus, and the memory controller will use this address to access the specific memory location.

Types of Bus



Example Scenario

1.CPU Request: The CPU needs to read data from memory location 0x1A3F.

2.Address Bus: The CPU places the address 0x1A3F on the address bus.

3.Memory Controller: The memory controller receives the address and accesses the memory location 0x1A3F.

4.Data Bus: The data from memory location 0x1A3F is placed on the data bus and sent back to the CPU.

Visual Representation

Imagine the address bus as a highway with multiple lanes (wires). Each lane can carry a bit of the address. The more lanes (wider bus), the more addresses it can carry simultaneously, allowing the CPU to access a larger memory space.

Types of Bus



d. Data bus

A **data bus** is another crucial part of a computer's bus system. It is responsible for transferring actual data between the CPU, memory, and other peripherals.

Types of Bus



Here are some key points about the data bus:

- **Width:** The width of the data bus (measured in bits) determines how much data can be transferred at one time. Common widths include 8-bit, 16-bit, 32-bit, and 64-bit.
- **Bidirectional:** The data bus is typically bidirectional, meaning data can flow in both directions—between the CPU and memory or between the CPU and input/output devices.
- **Speed:** The speed of the data bus affects the overall performance of the computer, as it determines how quickly data can be moved around the system.

In summary, the data bus plays a vital role in the overall performance and efficiency of a computer by facilitating the transfer of data between its various components.

Types of Bus



2. External bus

Known as expansion bus, allows peripherals to connect and communicate with the motherboard and its components.

BUS STRUCTURE

